

InSight3D Quality Inspection Report

AI-Based Internal Defect Segmentation in 3D-Printed Components

Component ID:	COMP-3D-ELEX-902	Inspection Date:	2026-09-05 20:12:25
Model Architecture:	U-Net (Keras)	Quality Status:	FAIL

Defect Analysis Summary Metrics

Metric Parameter	Measured Value	Unit / Status
Total Defect Count	159	Defect regions
Total Defect Area	9548.50	Pixels ²
Largest Defect Region	2949.00	Pixels ²
Defect Percentage	14.57%	Area %
Final Assessment	FAIL	Threshold: <1% Pass, 1-3% Warn

Detected Defect Inventory Breakdown

Defect #	Area (px ²)	Centroid (X, Y)	Bounding Box [X, Y, W, H]
1	92.0	(117, 238)	[103, 236, 29, 9]
2	98.0	(144, 240)	[134, 235, 22, 13]
3	16.0	(159, 234)	[155, 232, 8, 6]
4	4.0	(114, 232)	[112, 232, 5, 2]
5	4.0	(85, 231)	[84, 230, 3, 4]
6	3.5	(172, 228)	[169, 227, 7, 4]
7	12.0	(85, 226)	[81, 226, 10, 3]
8	20.5	(32, 225)	[30, 222, 6, 8]
9	13.0	(116, 224)	[114, 221, 6, 8]
10	9.5	(89, 222)	[87, 221, 6, 4]